

3.1 A generalization

To further illustrate how the method works let us try to find an approximation to $f'(x)$ again by using 3 points, without choosing the third one at the special position $x - h$. We keep the position of the three points general by choosing them as x , $x + h_1$ and $x + h_2$. If $h_1 < 0$ the second point is to the left of x , while it is to the right if $h_1 > 0$; the same is true for the second point. Of course we should expect that, if we choose $h_2 = -h_1$, we would recover the previous result (3.7).

Again, what we need to do is to find the right coefficients of the linear combination

$$\alpha f(x) + \beta f(x + h_1) + \gamma f(x + h_2), \quad (3.8)$$

that will produce such an improved approximation. In the previous case there was only one h and the mathematical definition of derivative suggested that it was efficient (in the sense of simplifying the algebra) to divide the linear combination by h as in (3.4). Now we have two h 's and it is not obvious what to use but, since we have a systematic method, we should be confident that it will lead us to the right result irrespective of whether we include a denominator with h_1 , h_2 or anything else. Our problem then is to find α , β and γ in such a way that the relation

$$\alpha f(x) + \beta f(x + h_1) + \gamma f(x + h_2) \simeq f'(x), \quad (3.9)$$

holds with an error (or remainder) having as high a power of h_1 and h_2 as possible.

For this purpose we use the Taylor series on the second and third terms writing

$$f(x + h_1) = f(x) + h_1 f'(x) + \frac{1}{2} h_1^2 f''(x) + \frac{1}{3!} h_1^3 f'''(x) + \dots, \quad (3.10)$$

$$f(x + h_2) = f(x) + h_2 f'(x) + \frac{1}{2} h_2^2 f''(x) + \frac{1}{3!} h_2^3 f'''(x) + \dots. \quad (3.11)$$

Upon substitution into the left-hand side of (3.9) we have

$$\begin{aligned} \alpha f(x) + & \beta \left[f(x) + h_1 f'(x) + \frac{1}{2} h_1^2 f''(x) + \frac{1}{3!} h_1^3 f'''(x) + \dots \right] + \\ & \gamma \left[f(x) + h_2 f'(x) + \frac{1}{2} h_2^2 f''(x) + \frac{1}{3!} h_2^3 f'''(x) + \dots \right] \simeq f'(x). \end{aligned} \quad (3.12)$$

Collecting terms we find

$$(\alpha + \beta + \gamma) f(x) + (\beta h_1 + \gamma h_2) f'(x) + \frac{1}{2} (\beta h_1^2 + \gamma h_2^2) f''(x) + \frac{1}{3!} (\beta h_1^3 + \gamma h_2^3) f'''(x) + \dots \simeq f'(x). \quad (3.13)$$

Upon comparing left- and right-hand sides we see that, if we choose α , β and γ in such a way that

$$\alpha + \beta + \gamma = 0, \quad (3.14)$$

$$\beta h_1 + \gamma h_2 = 1, \quad (3.15)$$

$$\frac{1}{2} (\beta h_1^2 + \gamma h_2^2) = 0, \quad (3.16)$$

$$\frac{1}{3!} (\beta h_1^3 + \gamma h_2^3) = 0, \quad (3.17)$$

the left-hand side would approximate $f'(x)$ up to terms containing the fourth and higher powers of the increments h_1 and h_2 . Since we only have three parameters, in general we can only satisfy three of these four conditions. Which ones to choose becomes evident when we realize that, in order to have a local approximation to $f'(x)$, the points $x + h_1$ and $x + h_2$ must be chosen close to x , so that the increments will be small (in a suitable sense to be specified later) and their powers will get smaller and smaller with

increasing order. For this reason we choose to satisfy the first three equations of the system since the powers h_1^3 and h_2^3 will be smaller than h_1^2 and h_2^2 . Satisfying the fourth one would require adding another parameter, which can be achieved by considering f at a fourth point $x + h_3$, and so on if we wish to remove still higher powers of the increments. With three points only, the best that we can do (at least in general) is to reduce the difference between the two sides of (3.9) to terms containing the third powers of the increments.

The system constituted by the first three equations of (3.15) is readily solved to find

$$\alpha = -\frac{h_1 + h_2}{h_1 h_2}, \quad \beta = \frac{h_2}{h_1(h_2 - h_1)}, \quad \gamma = -\frac{h_1}{h_2(h_2 - h_1)} \quad (3.18)$$

so that we have the general formula

$$f'(x) \simeq -\frac{h_1 + h_2}{h_1 h_2} f(x) + \frac{h_2}{h_1(h_2 - h_1)} f(x + h_1) - \frac{h_1}{h_2(h_2 - h_1)} f(x + h_2). \quad (3.19)$$

The coefficient of the third derivative f''' is readily found to be

$$-\frac{1}{3!} h_1 h_2 \quad (3.20)$$

so that no choice of h_1 and/or h_2 (other than, trivially, 0) can cause it to vanish. The difference between the two sides of (3.9) is then, to leading order,

$$\alpha f(x) + \beta f(x + h_1) + \gamma f(x + h_2) - f'(x) = -\frac{1}{3!} h_1 h_2 f'''(x) + \dots \quad (3.21)$$

This relation shows that the general formula (3.19) is accurate to second order: h_1 and h_2 are both of first order, and their product $h_1 h_2$ is therefore of second order.

Let us now look at some special cases

- Take $h_2 = h$, $h_1 = -h$; then we find

$$\alpha = 0, \quad \beta = -\frac{1}{2h}, \quad \gamma = \frac{1}{2h}, \quad (3.22)$$

so that

$$f'(x) = \frac{f(x + h) - f(x - h)}{2h} + O(h^2). \quad (3.23)$$

In this case the two points $x + h_1$ and $x + h_2$ happen to be on either side of x and equally distant from it, which justifies the denomination *centered difference* given to this formula. A point worth of notice is that, by using two points only, in general we end up with a formula accurate to first order in h as (1.24) shows. However, by choosing the two points at $x + h$ and $x - h$, we have an improved error of order h^2 . This is the remarkable property of the centered difference formula. We have already noted in the previous section that (3.23) can also be deduced by taking the average of the forward and backward formulae (1.25) and (1.26).

- Take $h_1 = h$, $h_2 = 2h$ to find a formula for the one-sided approximation from the right (also called *forward formula*, because it uses points ahead of the point of interest x):

$$f'(x) = \frac{4f(x + h) - 3f(x) - f(x + 2h)}{2h} + O(h^2). \quad (3.24)$$

- If $h_1 = -h$, $h_2 = -2h$ we have the formula for the one-sided approximation from the left (also called *backward formula*, because it uses points behind the point of interest x):

$$f'(x) = \frac{3f(x) - 4f(x - h) + f(x - 2h)}{2h} + O(h^2). \quad (3.25)$$

4 Higher-order derivatives

The general approach outlined in the previous section can also be applied to higher-order derivatives. Let us work out the second derivative, for example, for the special case in which we consider the point x and the two points $x \pm h$. As in (3.9) we write

$$\alpha f(x) + \beta f(x+h) + \gamma f(x-h) \simeq f''(x), \quad (4.1)$$

and, proceeding as in (3.12), we have

$$\begin{aligned} \alpha f(x) + \beta \left[f(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + \frac{1}{3!}h^3 f''' + \dots \right] + \\ \gamma \left[f(x) - hf'(x) + \frac{1}{2}h^2 f''(x) - \frac{1}{3!}h^3 f''' + \dots \right] \simeq f''(x), \end{aligned} \quad (4.2)$$

or, upon collecting pieces,

$$(\alpha + \beta + \gamma)f(x) + h(\beta - \gamma)f'(x) + \frac{h^2}{2}(\beta + \gamma)f''(x) + \frac{h^3}{3!}(\beta - \gamma)f''' + \dots \simeq f''(x). \quad (4.3)$$

In order for the left-hand side to equal the right-hand side it is necessary that

$$\alpha + \beta + \gamma = 0 \quad (4.4)$$

$$h(\beta - \gamma) = 0 \quad (4.5)$$

$$\frac{h^2}{2}(\beta + \gamma) = 1 \quad (4.6)$$

Upon solving this system we find

$$\alpha = -\frac{2}{h^2}, \quad \beta = \gamma = \frac{1}{h^2} \quad (4.7)$$

so that

$$f''(x) = \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} + O(h^2). \quad (4.8)$$

We see that in this special case something remarkable has happened: normally, with 3 points, we can only have 3 equations. Killing the coefficient of the term in f''' would amount to a fourth equation which normally⁶ would not be satisfied. But, with this special choice of the three points, this fourth equation is identical to the second one and is therefore “accidentally” satisfied! Note that, while for the first derivative we need a minimum of 2 points and, adding a third one, buys us more accuracy, there would be no way to estimate the second derivative with fewer than 3 points because we have three requirements: “knocking out” the terms containing f and f' in the left-hand side of (4.3) and be left with a coefficient 1 for the term f'' . There is no way we can satisfy these three requirements with fewer than three coefficients, and this requires the use of three points minimum. However, just as in the case of the first derivative, we can obtain a formula with better accuracy by adding points.

There is another way in which we can deduce (4.8) which is interesting. The second derivative is the derivative of the first derivative and, therefore, we can apply (1.31); we do so by using $h/2$ rather than h . In this way we find

$$f''(x) \simeq \frac{f'(x+h/2) - f'(x-h/2)}{2(h/2)}. \quad (4.9)$$

⁶“Normally” means that, if in place of $x \pm h$ we had taken two general points $x + h_1$ and $x + h_2$, the coefficient of f''' would not vanish.

Now we apply again (1.31), this time to approximate $f'(x \pm h/2)$:

$$f'(x + h/2) \simeq \frac{f([x + h/2] + h/2) - f([x + h/2] - h/2)}{2(h/2)} = \frac{f(x + h) - f(x)}{h} \quad (4.10)$$

$$f'(x - h/2) \simeq \frac{f([x - h/2] + h/2) - f([x - h/2] - h/2)}{2(h/2)} = \frac{f(x) - f(x - h)}{h} \quad (4.11)$$

Upon substituting into (4.9) we find

$$f''(x) \simeq \frac{1}{h} \left[\frac{f(x + h) - f(x)}{h} - \frac{f(x) - f(x - h)}{h} \right] \quad (4.12)$$

which is identical to (4.8).

How many points would we need to approximate, for example, the third derivative? When we carry out a Taylor series expansion of $f(x + h)$, for example, we have to write down three terms before we see f''' . By using the same approach as before, we need to kill the coefficients of $f(x)$, $f'(x)$ and $f''(x)$ and set equal to 1 the coefficient of f''' . Thus we need to impose 4 conditions, and this (except for possible special cases) would require 4 points. As in the previous cases, we might buy some better accuracy by adding more points to the formula but, at a minimum, we would need the value of $f(x)$ at four points. By the same argument we would need a minimum of five points for the fourth derivative, and so on.